

Chapter

3

Newtons Law of Motion

DEFINITION:

Dynamics is the branch of physics or mechanics that deals with the study of the motion of bodies in relation to the forces acting upon them.

- **State of Rest :**

"A body is said to be in rest if it is not changing its position with respect to its surroundings"

- **State of Motion:**

"A body is said to be in motion if it is changing its position with respect to its Surroundings"

NEWTON'S LAW OF MOTION:

In 1686, Sir Isaac Newton presented his three laws of motion. Newton studied motion of bodies and formulated the following three important laws of motion. The laws are shown below:

NEWTON'S FIRST LAW OF MOTION**Introduction**

In this law Newton explain the two important definitions first is the force & the second one is inertia.

Statement

A body remains at rest or continues to move with uniform velocity unless; it is acted upon by an unbalanced force.

Explanation

From Newton's first law of motion, we can conclude that the law consists of two parts: the first part states that a body will remain at rest or in uniform motion in a straight line unless acted upon by an unbalanced force.

Example

This law can also be explained with the help of following examples

1. A book lying on a table will remain in the same position indefinitely unless an external force, such as someone removing it, acts upon it.
2. A bullet is fired from a gun. Its motion is opposed both by air resistance & the pull of earth. If the pull of the earth and the air resistance could be eliminated, the bullet could go on moving in a straight line forever.

The second part of this law gives us the qualitative definition of the net force, which is stated as follows:

Definition of Force On The Basis Of First Law of Motion

"Force is an agent that produces or tends to produce a change in the state of rest or uniform motion of an object, i.e., it causes the object to accelerate".

First Law of Motion is Also Called Law of Inertia

First law of motion is also called law of inertia, because it points towards a very important property of matter INERTIA.

Definition of INERTIA

Inertia is that property of matter by virtue of which if it is in state of rest or motion it tries to remain in that state. OR Inertia is the tendency of an object resists a change in its state.

Experiments show that the inertia of an object is directly proportional to the mass of the object, i.e. the greater the mass of an object, greater will be the inertia.

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NEWTON'S SECOND LAW OF MOTION

Introduction

In this law of motion Newton provide a means for the quantitative measurement of force as well as mass.

Statement

"When a force acts on an object, it produces acceleration in its own direction, which is directly proportional to the magnitude of the force and inversely proportional to the mass of the object".

Explanation

If we push a body harder, it move faster. It velocity changes in the direction of the force exerted. From such experiences it is established that when a force acts upon a body, the acceleration produced is directly proportional to the force and inversely proportional to mass, symbolically it can be expressed as:

$$a \propto F \quad (\text{If mass is kept constant})$$

$$a \propto \frac{1}{m} \quad (\text{If force is kept constant})$$

Combining both equations we get:

$$a \propto F/m$$

$$a = k \frac{F}{m} \quad \text{where } k = 1$$

Where "F" is a (vector) sum of all the forces acting on the body, and 'm' is the mass of the mathematical expression of Newton's second law of motion.

$$\vec{a} = \frac{\vec{F}}{m}$$

In Vector Form:

$$\vec{F} = m\vec{a}$$

Second law of motion shows that the force is a product of mass and acceleration.

Note: This law provides quantitative determination of force

Unit

In M.K.S system, its unit is Newton (N) = (kg.m/s²).

In C.G.S system its unit is Dyne = (gm.cm/s²).

In F.P.S System its unit is ft-lb/s².

NEWTON'S THIRD LAW OF MOTION

Introduction

In this law Newton explains the action and reaction of the force. It is stated as follows.

Statement

To every action, there is an equal and opposite reaction.

Explanation

When a body "A" exerts a force on other body "B", it is called the action of force "A" on "B". The body "B" will also exerts a force on body "A", which will be equal in magnitude, but opposite in direction. this force is called the reaction of "B" on "A".

The force of body "A" on body "B" is written as $\vec{F}_{AB}$ and the force of body "B" on the body "A" is written as $\vec{F}_{BA}$, which be equal in magnitude but opposite in direction and these force lie on the line joining the two bodies. Symbolically, it can be expressed as

$$\vec{F}_{\text{action}} = -\vec{F}_{\text{reaction}}$$

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

Where negative sign shows that the two forces are acting in opposite's direction.

PROPERTY OF FORCE

The 3rd law of motion specifies the property of a force. In nature force is always occurs in pairs called action and reaction.

MOMENTUM:**Definition**

"The quantity of motion, which increases with the increase of mass and as well as of velocity and decreases with the decreases of mass as well as of velocity, is called momentum".

OR

"The product of mass and velocity is known as momentum".

Mathematical Expression

Mathematically it can be defined as the product of mass and velocity.

$$\vec{P} = m \vec{v}$$

Unit

In M.K.S system, its unit is N-S.

In C.G.S system its unit is Dyne-Sec.

In F.P.S System its unit is Lb-Sec.

Derivation Of The Unit Of Momentum

As momentum is the product of mass and velocity, so its unit is derived as follows.

$$\begin{aligned} \text{Momentum} &= \text{mass} \cdot \text{velocity} \\ &= \text{kg} \cdot \text{m/s} \\ &= \text{kg} \cdot \text{m/s}^2 \times \text{s} = \text{N-s} \end{aligned}$$

Therefore the unit of momentum is N - s.

Force In terms of Momentum:

According to the second law of motion

$$F = ma \quad \text{Eq - (1)}$$

$$\text{But } a = \frac{v_f - v_i}{t}$$

But

So equation (1) becomes

$$F = m \frac{v_f - v_i}{t}$$

$$F = \frac{mv_f - mv_i}{t} \quad \text{Eq - (2)}$$

As we know that

$$P = mv, \quad P_f = mv_f \text{ and } P_i = mv_i$$

So equation (2) becomes

$$\begin{aligned} F &= \frac{P_f - P_i}{t} \\ F &= \frac{\Delta P}{t} \end{aligned}$$

So it is proved that force is rate of change of momentum.

IMPULSE:

Impulse is defined as the change in momentum of an object when a force is applied over a period of time. It is equal to the product of the force and the time duration during which the force acts.

Mathematically, impulse ($\vec{J}$) is given

$$\vec{J} = \vec{F} \cdot \Delta t = \Delta \vec{P}$$

$$\vec{J} = \Delta \vec{P}$$

Unit

It is equal to the change of momentum, so it has same unit as momentum (N-s).

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Law of Conservation Of Linear Momentum

LAW OF CONSERVATION OF LINEAR MOMENTUM**Statement**

The momentum of an isolated system always remains constant.

OR

If there is no external force applied to a system, then the total momentum of that system remains constant.

Proof :

Consider a body of mass "m" and acceleration "a". then according to second law of motion.

$$\vec{F} = \frac{\Delta \vec{P}}{t}$$

If no external force acts on it then

$$F = \frac{\Delta P}{t} = 0$$

$$\Delta P = \text{change in momentum} = 0$$

Derivation:

Consider a system consisting of two bodies A & B of masses m_1 & m_2 respectively. These are moving in a straight line, with velocities u_1 and u_2 before collision. On colliding with each other, their final velocities will be v_1 & v_2 respectively.

Upon Collision m_1 exert force on m_2 :

$$F_{12} = (m_2 v_2 - m_2 u_2) / \Delta t$$

Upon Collision m_2 exert force on m_1 :

$$F_{21} = (m_1 v_1 - m_1 u_1) / \Delta t$$

According to Newton's Third Law:

$$F_{12} = -F_{21}$$

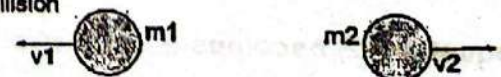
Before collision



During collision



After collision



$$\frac{(m_2 v_2 - m_2 u_2)}{\Delta t} = - \frac{(m_1 v_1 - m_1 u_1)}{\Delta t}$$

$$m_1 v_1 + m_2 v_2 = m_1 u_1 + m_2 u_2$$

Therefore:

Total Momentum Before Collision = Total Momentum After Collision

This is known as law of conservation of momentum. Thus the above equation shows that the total momentum of the system before and after collision is the same. The mutual action and reaction of the bodies of an isolate system are unable to change the momentum of the system i.e. the momentum of the system is conserved.

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Elastic Collision

ELASTIC COLLISION

An elastic collision is that in which the momentum of the system as well as the kinetic energy of the system before and after collision is conserved i.e. remains same.

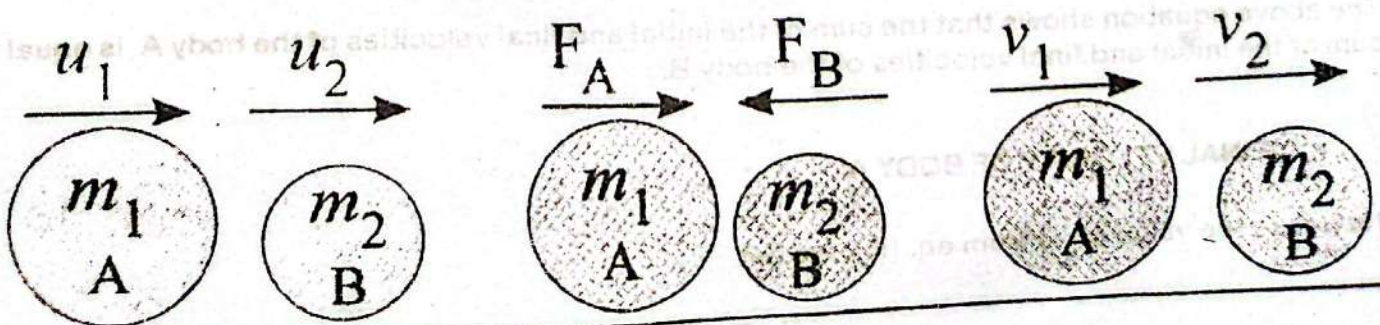
INELASTIC COLLISION

In an inelastic collision, the momentum of the system before and after the collision is conserved. However, the kinetic energy changes, i.e the total kinetic energy is not constant.

ELASTIC COLLISION IN ONE DIMENTION:

Consider two smooth non-rotating spheres A and B of masses m_1 and m_2 respectively, moving initially along the line joining their centers with velocities u_1 and u_2 . If u_1 is greater than u_2 , so they collide with one another and after having an elastic collision start moving with velocities v_1 and v_2 respectively in the same line and direction.

Now the momentum of the system before collision = $m_1 u_1 + m_2 u_2$
 And the momentum of the system after collision = $m_1 v_1 + m_2 v_2$



According to law of conservation of momentum, we have

Total momentum before collision = Total momentum after collision

$$\begin{aligned} m_1 u_1 + m_2 u_2 &= m_1 v_1 + m_2 v_2 \\ m_1 u_1 - m_1 v_1 &= +m_2 v_2 - m_2 u_2 \\ m_1 (u_1 - v_1) &= m_2 (v_2 - u_2) \end{aligned} \quad (i)$$

$$\text{K.E of the system before collision} = \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2$$

$$\text{K.E of the system after collision} = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

As the collision is elastic, so the kinetic energy of the system is also conserved. K.E of the system before collision = K.E. of the system after collision.

K.E before collision = K.E after collision

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$$\frac{1}{2} (m_1 u_1^2 + m_2 u_2^2) = \frac{1}{2} (m_1 v_1^2 + m_2 v_2^2)$$

$$m_1 u_1^2 + m_2 u_2^2 = m_1 v_1^2 + m_2 v_2^2$$

$$m_1 u_1^2 - m_1 v_1^2 = m_2 v_2^2 - m_2 u_2^2$$

$$m_1 (u_1^2 - v_1^2) = m_2 (v_2^2 - u_2^2)$$

(ii)

Dividing Eq(II) by Eq(I) we get:

$$\begin{aligned} \frac{m_1 (u_1^2 - v_1^2)}{m_1 (u_1 - v_1)} &= \frac{m_2 (v_2^2 - u_2^2)}{m_2 (v_2 - u_2)} \\ \frac{u_1^2 - v_1^2}{u_1 - v_1} &= \frac{v_2^2 - u_2^2}{v_2 - u_2} \\ \frac{(u_1 - v_1)(u_1 + v_1)}{(u_1 - v_1)} &= \frac{(v_2 - u_2)(v_2 + u_2)}{(v_2 - u_2)} \end{aligned}$$

$$u_1 + v_1 = v_2 + u_2 \quad (iii)$$

The above equation shows that the sum of the initial and final velocities of the body A is equal to the sum of the initial and final velocities of the body B.

• FINAL VELOCITY OF BODY A

Now take the value of v_2 from eq. (iii), we get

$$v_2 = u_1 + v_1 - u_2$$

Put the value of v_2 in eq, (i)

$$\begin{aligned} m_1(u_1 - v_1) &= m_2(v_2 - u_2) \\ m_1(u_1 - v_1) &= m_2(u_1 + v_1 - u_2 - u_2) \\ m_1(u_1 - v_1) &= m_2(u_1 + v_1 - 2u_2) \\ m_1u_1 - m_1v_1 &= m_2u_1 + m_2v_1 - 2m_2u_2 \\ m_1u_1 - m_2u_1 + 2m_2u_2 &= m_1v_1 + m_2v_1 \end{aligned}$$

Or it can be written as

$$\begin{aligned} m_1v_1 + m_2v_1 &= m_1u_1 - m_2u_1 + 2m_2u_2 \\ v_1(m_1 + m_2) &= (m_1 - m_2)u_1 + 2m_2u_2 \\ v_1 &= \frac{(m_1 - m_2)u_1 + 2m_2u_2}{(m_1 + m_2)} \\ v_1 &= \left(\frac{m_1 - m_2}{m_1 + m_2} \right) u_1 + \left(\frac{2m_2}{m_1 + m_2} \right) u_2 \end{aligned}$$

• FINAL VELOCITY OF BODY B

Similarly we take the value of v_1 from equation (iii), we get

$$v_1 = v_2 + u_2 - u_1$$

Put this value of v_1 in equation (i) .

$$\begin{aligned} m_1(u_1 - v_1) &= m_2(v_2 - u_2) \\ m_1[u_1 - (v_2 + u_2 - u_1)] &= m_2(v_2 - u_2) \\ m_1(u_1 - v_2 - u_2 + u_1) &= m_2(v_2 - u_2) \\ m_1(2u_1 - v_2 - u_2) &= m_2(v_2 - u_2) \\ 2m_1u_1 - m_1v_2 - m_1u_2 &= m_2v_2 - m_2u_2 \end{aligned}$$

Or it can be written as

$$\begin{aligned} m_1v_2 + m_2v_2 &= 2m_1u_1 + m_2u_2 - m_1u_2 \\ v_2(m_1 + m_2) &= 2m_1u_1 + u_2(m_2 - m_1) \\ v_2 &= \frac{2m_1u_1 + u_2(m_2 - m_1)}{(m_1 + m_2)} \end{aligned}$$

Or it can be written as

$$\begin{aligned} v_2 &= \frac{2m_1u_1}{(m_1 + m_2)} + \frac{u_2(m_2 - m_1)}{(m_1 + m_2)} \\ v_2 &= \left(\frac{2m_1}{m_1 + m_2} \right) u_1 + \left(\frac{m_2 - m_1}{m_1 + m_2} \right) u_2 \end{aligned}$$

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Friction

FRICTION

Friction is the force that resists the sliding or movement between two surfaces in contact. It acts to slow down or stop their motion.

Coefficient Of Friction

The coefficient of friction is a number that represents how much frictional force exists between two surfaces in contact. It is the ratio of the frictional force to the normal force pressing the surfaces together. It is denoted by μ

$$\mu = \frac{F}{R}$$

Where

F=Limiting friction,

R=Normal reaction ,

 μ =Coefficient of friction**Unit**

It has no unit , therefore it is dimensionless.

Classical View of Friction VS Modern view of Friction:

- **Classical view:** Friction is caused by the interlocking of two rough surfaces.
- **Modern view:** Friction is caused by the intermolecular forces of attraction between two surfaces in contact.

Types of Friction:• **STATIC FRICTION**

Static friction occurs when the two objects are not moving relative to each other (like a book on a desk). The coefficient of static friction is typically denoted as μ_s . The initial force to get an object moving is often dominated by static friction. The static friction is higher than the kinetic friction.

$$f_s \leq \mu_s n$$

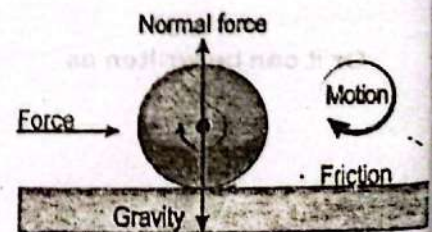
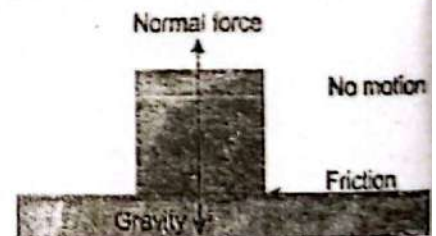
Example

Limiting friction is the maximum value of static friction, or the force of friction that acts when a body is just on the verge of motion on a surface.

• **KINETIC FRICTION**

Kinetic (or dynamic) friction occurs when two objects are moving relative to each other and rubs together (like a sled on the ground). The coefficient of kinetic friction is typically denoted as μ_k , and is usually less than the coefficient of static friction.

$$f_k = \mu_k n$$



Rolling Friction

Example

Sliding friction is when two objects are rubbing against each other. Putting a book flat on a desk and moving it around is an example of sliding friction

The Kinetic Friction has two types:

Rolling Friction: is the force that resists the rolling motion of an object on a surface.

Sliding Friction: is the force that resists the sliding motion of an object on a surface.

Rolling friction is less than sliding friction due to the reduced contact area and less resistance compared to sliding friction.



Sliding Friction

LAWS OF FRICTION:

1. Law of opposition: The frictional force always opposes the relative motion of two surfaces.
2. Law of parallelism: The frictional force is parallel to the surfaces in contact.
3. Law of proportionality: The frictional force is directly proportional to the normal reaction.
4. Law of independency of area: The frictional force is independent of the area of contact.
5. Law of nature: The frictional force depends upon the nature of the two surfaces in contact and their state of roughness.
6. Law of kinetic friction: The kinetic friction is independent of the relative velocities of the surfaces.

Angle Of Friction:

The angle of friction is defined as the angle between the direction of the resultant force (which is the combination of the normal reaction and the frictional force) and the direction of the normal reaction.

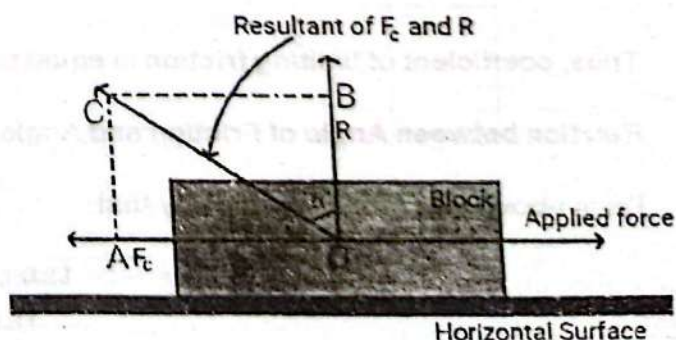
Consider a body lying on a horizontal surface. Let R be the normal reaction and F_c be the frictional force. If α is the angle made by the resultant of normal reaction and frictional force with the normal reaction, then, α is the angle of friction.

$$\tan \alpha = \frac{BC}{OB}$$

$$\tan \alpha = \frac{OA}{OB}$$

$$\tan \alpha = \frac{F_c}{R}$$

$$\therefore \tan \alpha = \mu$$



Therefore, coefficient of limiting friction is equal to the tangent of angle of friction

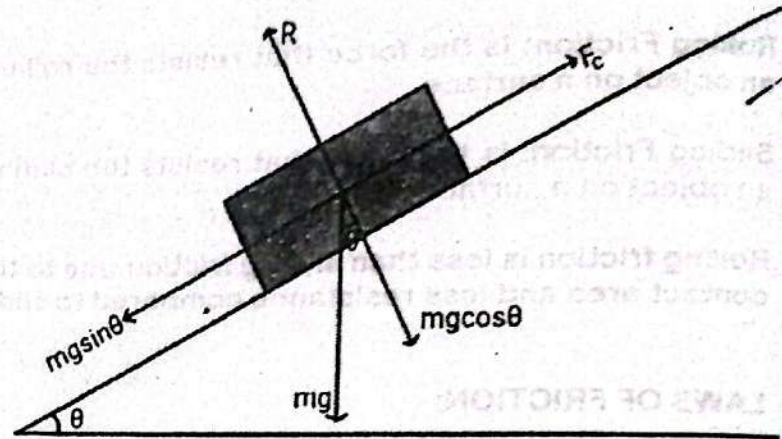
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Angle Of Repose:

Angle of repose is defined as the angle of an inclined plane, at which a body placed on it just begins to slide.

Place a body of mass m on an inclined plane. Increase the angle of inclined plane till the body just begins to slide. If θ is the angle of inclination at which the body just begins to slide, then θ is the angle of repose.

The weight mg of the body acts vertically downward can be resolved into two components; $mg \cos \theta$ and $mg \sin \theta$



If the body just begins to slide, then

1 $mg \sin \theta$ is equal to the frictional force

$$mg \sin \theta = F_c$$

2 $mg \cos \theta$ is equal to the normal reaction

$$mg \cos \theta = R$$

Dividing Equations we get

$$\frac{mg \sin \theta}{mg \cos \theta} = \frac{F_c}{R}$$

$$\tan \theta = \frac{F_c}{R}$$

$$\tan \theta = \mu$$

Thus, coefficient of limiting friction is equal to the tangent of the angle of repose.

Relation between Angle of Friction and Angle of Repose

From above Equations we can say that:

$$\tan \alpha = \mu = \tan \theta$$

$$\tan \alpha = \tan \theta$$

$$\alpha = \theta$$

Thus, angle of friction is equal to the angle of repose (In the limiting Friction).

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Multiple Choice Questions

MCQ'S		ANS.
1. The rate of change of linear momentum of a body is called a: a) Linear force c) Power b) Angular force d) Impulse		a
2. The term mass refers to the same physical concept as: a) weight c) force b) inertia d) acceleration		b
3. Which one of the following force is also called as self-adjusting force ? a) frictional force c) Weight b) tension d) Thrust		a
4. The laws of motion shows the relationship between: a) velocity and acceleration c) mass and acceleration b) mass and velocity d) force and acceleration		d
5. The motion of rocket in the space is according to the law of conservation of: a) Energy c) mass b) linear momentum d) angular momentum		b
6. A bomb of mass 12 kg initially at rest explodes into two pieces of masses 4kg and 8kg . The speed of 8kg mass is 6 m/s. The kinetic energy The kinetic energy of 4 kg mass is: a) 32 J c) 114 J b) 48 J d) 288 J		d
7. If momentum is increased by 20% then K.E increases by: a) 44% b) 66% b) 55% d) 77%		a
8. The kinetic energy of body of mass 2kg and momentum of 2 Ns is: a) 1 J c) 3 J b) 2 J d) 4 J		a
9. For the same kinetic energy, the momentum is maximum for: a) an electron c) a deuteron b) a proton d) alpha particle		d
10. A 3kg bowling ball experiences a net force of 15N. What will be its acceleration, a) 35 m/s^2 c) 5 m/s^2 b) 7 m/s^2 d) 35 m/s^2		c

***** Exam Practice MCQs *****

1. Angle of friction and angle of repose are
 - (a) Equal to each other
 - (b) Not equal to each other
 - (c) Proportional to each other
 - (d) None of the above
2. A player catches a ball of mass 150 gm moving at a rate of 20 m/s. If the process of catching is to be completed in 0.1 sec. What is the force exerted by the ball on the hands of the player?
 - (a) 3000 N
 - (b) 300 N
 - (c) 30 N
 - (d) 0.3 N
3. Which of the following is momentum closely related to?
 - (a) Force
 - (b) Power
 - (c) Impulse
 - (d) Kinetic Energy
4. When a force acts on a ball of mass 150 g for 0.1 sec, it produces an acceleration of 20 m/s². What is the value of this impulsive force?
 - (a) 0.5 N-s
 - (b) 0.1 N-s
 - (c) 0.3 N-s
 - (d) 1.2 N-s
5. The inherent property, with which a body resists any change in its state of motion is known as
 - (a) Force
 - (b) Momentum
 - (c) Inertia
 - (d) Acceleration
6. Which is the branch of physics that deals with the motion of a body by considering the cause?
 - (a) Statics
 - (b) Thermodynamics
 - (c) Dynamics
 - (d) Astronomy
7. The velocity of a body after elastic collision with a body of the same mass at rest will be
 - (a) Reduced to half
 - (b) Doubled
 - (c) Un-altered
 - (d) Zero
8. The ratio of limiting friction to the normal reaction is called
 - (a) Force of friction
 - (b) Co-efficient of friction
 - (c) Kinetic friction
 - (d) Static friction
9. In case of inelastic collision:
 - (a) Both momentum and K.E are conserved
 - (b) neither momentum nor K.E is conserved
 - (c) Only momentum is conserved
 - (d) Only K.E is conserved
10. When a body slides over a surface the kinetic friction (f_k) and the static friction (f_s) are related as
 - (a) $f_k < f_s$
 - (b) $f_s < f_k$
 - (c) $f_k = 0$
 - (d) $f_s = 0$
11. When a constant force is applied on a body it moves with
 - (a) Constant speed
 - (b) Constant velocity
 - (c) Constant acceleration
 - (d) Constant displacement

12. The rate of change of linear momentum is equal to (a) Acceleration (c) Force (b) Torque (d) Angular velocity	c
13. When a body is placed at an inclined frictionless plane the force by which the body slides down (a) $mg\cos\theta$ (c) $g\cos\theta$ (b) $mg\sin\theta$ (d) Zero	b.
14. The ratio of kinetic friction to limiting static friction is always (a) Zero (c) Less than one (b) One (d) Greater than one	c
15. The angle of the inclined plane at which a body just begins to slide down the plane is called (a) Angle of repose (c) Angle of Projection (b) Angle of Elevation (d) Angle of Incidence	a
16. A body of mass 'm' moving with velocity 'V' collides with a stationary body of mass '2m' and combine to form one body. The velocity of the combine body after collision, is: (a) 3 V (b) V/3 (c) 9V (d) 4 V	b
17. In CGS system the unit of linear momentum is called : (a) Dyne (c) Dyne.cm (b) Dyne.sec (d) Dyne/sec	c
18. Impulse is equal to: (a) Force x time (c) Change In Acceleration (b) Momentum x time (d) Change In Velocity	a
19. The modern view of friction takes into account the: (a) Roughness of the surfaces in contact (b) Adhesion between the surfaces in contact. (c) Temperature of the surfaces in contact (d) All of the above.	d
20. Which of the following is not a characteristic of static friction? (a) It is a self-adjusting force (b) It is proportional to the normal force. (c) It is always in the direction opposite to the motion of the object (d) It can be overcome by applying a sufficiently large force.	c
21. Which of the following is not an example of Newton's first law of motion? (a) A car moving at a constant speed on a straight road (b) A ball rolling down a hill (c) A book sitting on a table (d) A rocket taking off from the launch pad	d

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Chapter Formulae & Numericals

Force	$F = ma$
Momentum	$P = mv$
Relation between force and linear momentum	$F = \frac{\Delta P}{\Delta t}$
Impulse	$\vec{j} = F \cdot \Delta t$ $\vec{j} = \Delta P$
Law of conservation of linear momentum	$m_1 U_1 + m_2 U_2 = m_1 V_1 + m_2 V_2$
Elastic collision	$V_1 = \frac{(m_1 - m_2)}{(m_1 + m_2)} U_1 + \frac{2m_2 U_2}{m_1 + m_2}$ $V_2 = \frac{2m_1 U_1}{(m_1 + m_2)} + \frac{(m_1 - m_2) U_2}{(m_1 + m_2)}$
Coefficient of Friction	$\mu = \frac{F}{R}$